
EXPERIMENT – 02

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Aim

To install and configure Docker, pull Docker images, run containers, and manage the container lifecycle using Docker commands.

Objectives

- To pull Docker images from Docker Hub
- To run containers with port mapping
- To verify running containers
- To manage container lifecycle (start, stop, remove)

Theor y

Docker is an open-source containerization platform that allows applications to be packaged along with their dependencies into lightweight and portable containers. Containers run on a shared operating system kernel, making them faster and more resource-efficient than traditional Virtual Machines.

A Docker Image is a read-only template used to create containers.

A Docker Container is a running instance of a Docker image.

Docker follows a client-server architecture, where:

- The Docker Client sends commands
- The Docker Daemon executes them
- Docker Hub stores container images

Software Requirements

- Windows Operating System
- Docker Desktop (with WSL integration)
- Ubuntu (WSL distribution)

Procedure / Steps to Perform the Experiment

Step 1: Pull Docker Image

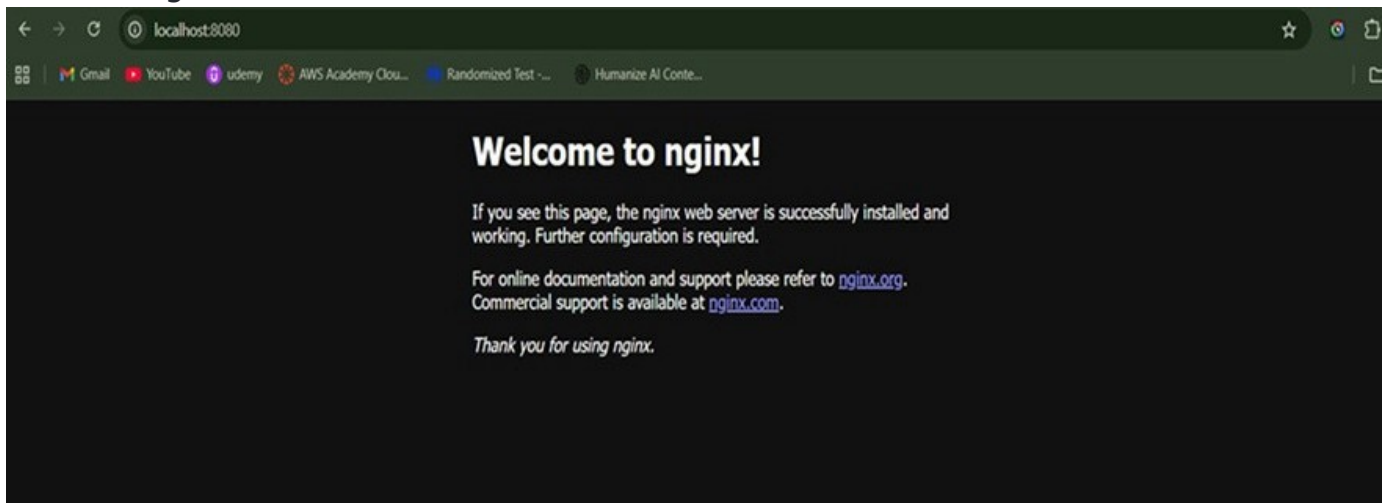
Pull the Nginx image from Docker Hub:

```
docker pull nginx
```

```
aryanraj@DESKTOP-DU3B1ES:/mnt/c/Users/Hp$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
46bf3a120c8e: Pull complete
bae5a1799a80: Pull complete
47cd406a84ef: Pull complete
0c8d55a45c0d: Pull complete
2e02dba24409: Download complete
a5d78d617315: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Step 2: Run Container with Port Mapping

Run the Nginx container in detached mode:



```
docker run -d -p 8080:80 nginx
```

Explanation:

- d → Runs container in background (detached mode)
- p 8080:80 → Maps host port 8080 to container port 80
- nginx → Docker image name

Step 3: Verify Running Containers

To check running containers:

```
docker ps
```

```
aryanraj@DESKTOP-DU3B1ES:/mnt/c/Users/Hp$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
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This command displays:

- Container ID
- Image name
- Container Port mapping

```
aryanraj@DESKTOP-DU3B1ES:/mnt/c/Users/Hp$ docker stop nginx-container
```

Images [Give feedback](#)

Local

My Hub

16.38 KB / 0 Bytes in use 2 Images

Last refresh: 14 hours ago

☐	Name	Tag	Image ID	Created	Size	Actions
☐	hello-world	latest	ef54e839ef54	6 months ago	25.9 KB	Run More Delete
☐	nginx	latest	341bf0f3ce6c	13 days ago	239.91 MB	Run More Delete

Showing 2 items

Walkthroughs

FROM node

RUN mkdir -p

WORKDIR /app

COPY package

How do I run a container?

6 mins

docker hub image

Run Docker Hub images

5 mins

[View more in the Learning center](#)

RAM 1.07 GB CPU 1.76% Disk: 1.71 GB used (limit 1006.85 GB)

[Terminal](#) [Update available](#)

Images / nginx:latest

nginx:latest

341bf0f3ce6c

CREATED

13 days ago

SIZE

239.91 MB

Recommended fixes

Run

[Delete](#)

Analyzed by [docker scout](#)

Layers (18)

Vulnerabilities

Packages

Give feedback

0	# debian.sh --arch 'amd64' out/ 'trixie' '@1769990400'	87.43 MB
1	LABEL maintainer=NGINX Docker Maintainers <docker-m...	0 B
2	ENV NGINX_VERSION=1.29.5	0 B
3	ENV NJS_VERSION=0.9.5	0 B
4	ENV NJS_RELEASE=1~trixie	0 B
5	ENV ACME_VERSION=0.3.1	0 B
6	ENV PKG_RELEASE=1~trixie	0 B
7	ENV DYNPKG_RELEASE=1~trixie	0 B
8	RUN /bin/sh -c set -x && groupadd --system --gid 101 ngin...	86.65 MB
9	COPY docker-entrypoint.sh / # buildkit	8.19 KB
10	COPY 10-listen-on-ipv6-by-default.sh /docker-entrypoint.d...	12.29 KB

This image has not been analyzed

You can use Docker Scout to analyze local images and list its vulnerabilities.

Start analysis

Enable background indexing in Settings so your results are always ready.

RAM 1.02 GB CPU 0.12% Disk: 1.71 GB used (limit 1006.85 GB)

[Terminal](#) [Update available](#)

Step 4: Stop and Remove Container

To stop the running

container: `docker stop`

<container_id> To remove

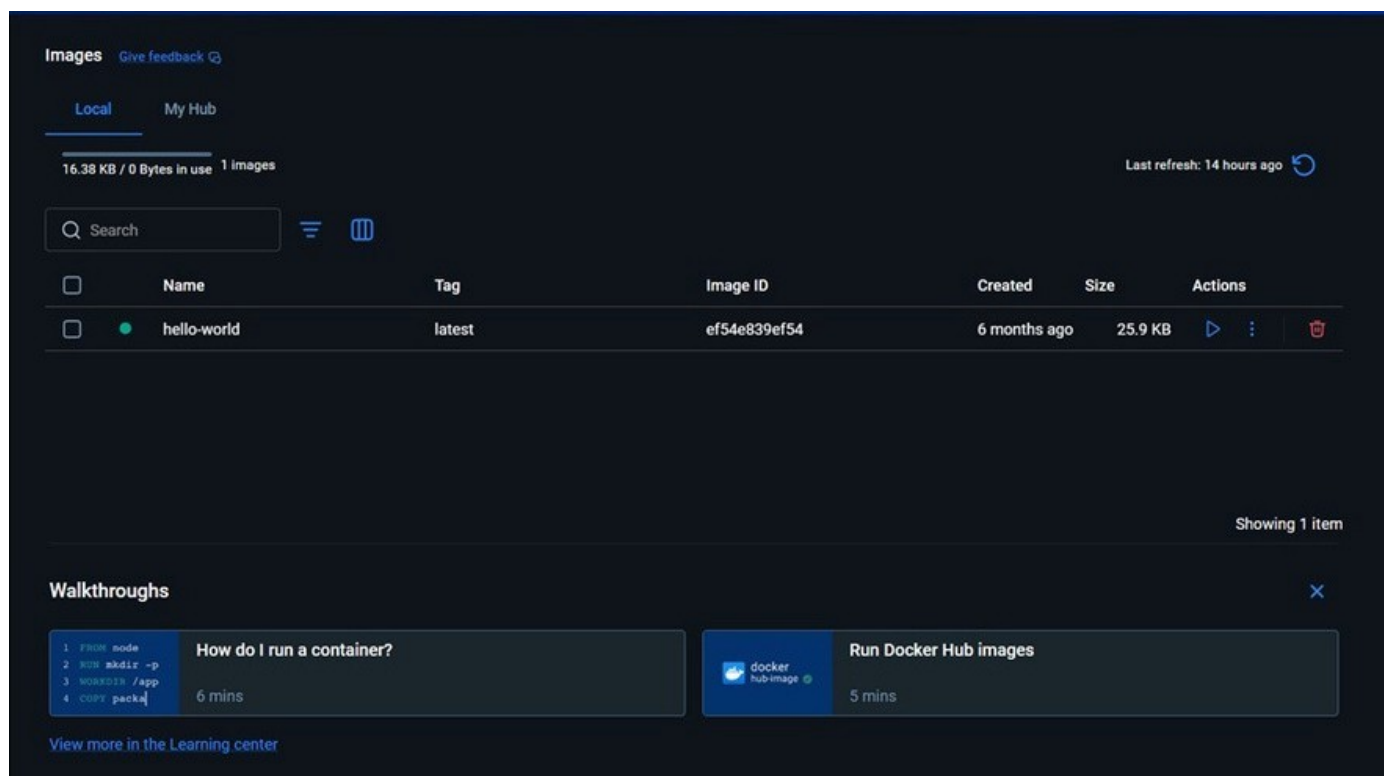
the container:

```
docker rm <container_id>
```

Step 5: Remove Docker Image

To remove the downloaded image:

```
docker rmi nginx
```



This frees disk space by deleting the unused image.

Result

Docker images were successfully pulled.

Containers were executed using port mapping.

Container lifecycle management commands (start, stop, remove) were performed successfully.